

FTEC5660 Assignment 2 part 2

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1. Agent Design and Architecture

This agentic system is designed to autonomously interact with Moltbook, a lightweight social discussion platform. The core architecture is built upon the LangChain framework and powered by a Large Language Model (LLM), utilizing a ReAct (Reasoning and Acting) mechanism to iterate through observations, thoughts, and actions.

Core Components:

- **LLM Engine:** The system utilizes the gemini-2.5-flash model via Google Vertex AI. The temperature is explicitly set to 0 to ensure deterministic outputs and strict adherence to system rules.
- **Agent Loop:** The main execution loop (`moltbook_agent_loop`) is configured with a maximum of 8 turns. This provides the agent with sufficient iterations to handle multi-step tasks, such as reading documentation, processing it, and executing subsequent actions.
- **Tool Set:** The agent's toolset is divided into two categories to maximize adaptability:
 1. **Static Tools:** Hardcoded functions for standard operations like `get_feed`, `search_moltbook`, `create_post`, `comment_post`, and `upvote_post`.
 2. **Adaptive Tool Execution (Documentation-Driven Architecture):** To handle newly required tasks such as community subscriptions, the agent is equipped with `read_api_docs` to autonomously fetch and parse the platform's API documentation (`skill.md`). Instead of relying purely on rigid assumptions, the agent learns the required parameters and community guidelines first, and then invokes the targeted `subscribe_submolt` tool to execute the authenticated HTTP request.

2. Decision Logic and Autonomy Level

This agent demonstrates a high level of autonomy by possessing the capability to "read and learn" rather than relying strictly on pre-programmed endpoints.

Decision Logic and Execution Chain:

1. **Authentication:** The agent automatically includes the registered Moltbook API key to authenticate with the Moltbook API.
2. **Autonomous API Discovery:** When instructed to subscribe to the submolt /m/ftec5660, the agent recognizes the absence of a dedicated subscribe tool. Following its System Prompt, it first calls read_api_docs to parse the API documentation.
3. **Informed Execution:** After the LLM autonomously extracts the correct platform rules and formatting constraints from the documentation, it intelligently processes the target submolt string (e.g., stripping the /m/ prefix as required) and utilizes the subscribe_submolt tool to successfully complete the subscription to /m/ftec5660.
4. **Targeted Social Interaction:** The agent accurately targets the required post (<https://www.moltbook.com/post/47ff50f3-8255-4dee-87f4-2c3637c7351c>). It first calls upvote_post to register an upvote. Subsequently, utilizing its LLM reasoning, it formulates a professional, contextually relevant comment and publishes it via comment_post.

3. Screenshots/Logs of Moltbook Interactions

Execution Log:

```
[08:30:08] [TOOL] [1] Calling 'upvote_post'
[08:30:08] [TOOL. ARGS] {
  "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c"
}
[08:30:09] [TOOL. RESULT] upvote_post finished (success) in 0.35s
[08:30:09] [TOOL. OUTPUT] {
  "success": true,
  "message": "Upvoted! 🍷",
  "action": "upvoted",
  "author": {
    "name": "BaoNguyen"
  },
  "already_following": false,
  "tip": "Upvotes are free and they mean a lot. Keep it up! 🍷"
}
```

```
[08:30:11] [TOOL. RESULT] comment_post finished (success) in 0.48s
[08:30:11] [TOOL. OUTPUT] {
  "success": true,
  "message": "Comment added! 🍷",
  "comment": {
    "id": "64e79039-ebf5-4077-ae46-9b6e686149db",
    "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c",
    "content": "The true power of AI/ML lies not just in automation, but in its capacity",
    "author_id": "79bcf56f-3446-4ca7-954a-c6c3c22d56dc",
    "author": {
      "id": "79bcf56f-3446-4ca7-954a-c6c3c22d56dc",
      "name": "5x1.68034214",
      "description": "va",
      "avatarUrl": null,
      "banned": 0,
      "followerCount": 0,
      "followingCount": 0,
      "isClaimed": true,
      "isActive": true,
      "createdAt": "2026-02-27T07:57:13.934Z",
      "lastActive": null
    }
  }
}
```

```
"type": "tool_call"
}
[08:30:06] [TOOL] [1] Calling 'subscribe_submolt'
[08:30:06] [TOOL. ARGS] {
  "submolt_name": "ftec5660"
}
[08:30:07] [TOOL. RESULT] subscribe_submolt finished (success) in 1.09s
[08:30:07] [TOOL. OUTPUT] {
  "success": true,
  "message": "Subscribed to m/ftec5660! 🍷",
  "action": "subscribed"
}
[08:30:07] [TURN] Turn 2 completed in 2.69s
[08:30:07] [TURN] Turn 3/8 started
[08:30:08] [LLM] Model responded
[08:30:08] [LLM. CONTENT] <empty>
[08:30:08] [LLM. TOOL_CALLS] [
```

Moltbook Interaction Verification:

- **Target Post:** <https://www.moltbook.com/post/47ff50f3-8255-4dee-87f4->

2c3637c7351c

- The agent has successfully generated an upvote and comment on the specified post.

Screenshot of my agent's activity in moltbook:

